

AR Electric Field — Marker Encoding

This lab tracks printed **3×3 matrix-code (barcode)** markers with AR.js — no marker training needed. There are **two independent schemes**: a **Plane** scheme (flat cards on a desk, charge at the marker centre) and a **Cube** scheme (markers on a box, charge at the cube centre). Because you only ever run **one scheme at a time**, the two schemes deliberately **reuse the same barcode values** — code 0 means a positive charge in the Plane app but a face of the positive cube in the Cube app. Print at 100% (actual size) and keep the white quiet-zone border around every code.

Plane scheme — flat markers, charge AT the marker centre

Code	Marker	Role	What / where
0		Positive charge (+)	A positive point charge sits at the centre of this card. Field lines radiate outward.
1		Negative charge (−)	A negative point charge at the card centre. Field lines converge inward.
2		Positive electron gun (fires e^+)	Emits positive test charges along the printed arrow . They curve along the field.
3		Negative electron gun (fires e^-)	Emits negative test charges along the printed arrow .

Cube scheme — markers on a box, charge at the CUBE CENTRE

Each cube uses **five** distinct face markers (top + four sides). Every visible face reports its own pose; the app steps each one inward along its normal by $\frac{1}{2}$ the **cube edge** and averages them, so all faces resolve to the same point — the cube centre — and the charge never jumps as you circle the cube.

Code	Markers	Role	What / where
0–4	    	Positive cube — 5 faces	Positive charge at the centre of this box.
5–9	    	Negative cube — 5 faces	Negative charge at the centre of a second box.
10		Positive electron gun (fires e^+)	Lay flat by the cubes; fires e^+ along its printed arrow.
11		Negative electron gun (fires e^-)	Fires e^- along its printed arrow.

Electron guns & firing direction

Each scheme has **two** guns so you can fire **both signs of test charge**: a positive gun that fires e^+ (positive electrons / positrons) and a negative gun that fires e^- (electrons). The gun's **firing direction is the printed arrow** on its card, which echoes the marker's own orientation: the app fires along the marker's local +X axis, projected onto the desk. So

the beam always leaves the card the way the arrow points. If a beam comes out rotated from what you expect, add **?gundir=1** (values 0–3, each a quarter-turn) to the page URL. A positive test charge is pushed along the field; a negative one against it, so the two guns trace the field in opposite senses.

Printing & tracking tips

- Print at **100% / actual size** — plane markers 4 cm, cube face markers 3.2 cm.
- Keep the **white border** (quiet zone) around each code; don't crop tight.
- Use even lighting and a matte (non-glossy) surface.
- Bigger markers track from farther away.
- In the live apps the layout **auto-locks** once the markers are seen steadily, freezing the field shape; the camera pose is then recovered from the **average of all visible markers** so the scene stays glued to the desk without jitter.